

# Project

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November 2025

## 1 Introduction

### Dataset Gathering, Motivation, and Initial Pre-processing

#### Dataset Description

- Clearly describe the dataset you are using. Mention the name, source, and provide a working URL (e.g., from Kaggle, UCI, or other public repositories).
  - We used the Diabetes prediction dataset from Mohammed Mustafa, available on Kaggle.com. website<sup>1</sup>.
- Specify the dataset size: total samples and features. — **redcode**
  - The Diabetes prediction datasets consists of 100,000 samples and 8 features. We plan on having 80,000 training samples and 20,000 test samples as specified by the code.
- Describe the type of data (e.g., tabular, image, text, time-series).
  - Our data is tabular data which consists of sample (or patient) features such as blood glucose levels, gender, age, if they have hypertension, if they have heart disease, if they have a history of smoking, their BMI levels, and hemoglobin levels.
- Summarize key variables and their meanings (e.g., input features and labels). — **redcode**
  - Our samples each have 8 input features and one label. The input features are gender, age, hypertension, heart disease, smoking history, bmi, HbA1c level, and blood glucose level. The label is diabetes, 1 for "has diabetes" and 0 for "does not have diabetes". (FOR SOME REASON ITS NOT LETTING ME OUT THE FIGURES UNDER "Data Loading and Initial Pre-processing")

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<sup>1</sup><https://www.cs.toronto.edu/~kriz/cifar.html>

```
print(df.head())
```

|   | gender | age      | hypertension | heart_disease | smoking_history | bmi       |
|---|--------|----------|--------------|---------------|-----------------|-----------|
| 0 | 0      | 0.01947  | 1.692784     | -0.284439     | 0.936379        | 0.963327  |
| 1 | 0      | 0.01947  | 0.538886     | -0.284439     | -0.282578       | -1.153468 |
| 2 | 1      | 1.187234 | -0.616891    | -0.284439     | -0.282578       | 0.963327  |
| 3 | 0      | 0.01947  | -0.261399    | -0.284439     | -0.282578       | -0.624269 |
| 4 | 1      | 1.187234 | 1.515858     | 3.515687      | 0.936379        | -0.624269 |

  

|   | HbA1c_level | blood_glucose_level | diabetes |
|---|-------------|---------------------|----------|
| 0 | 1.081786    | 0.847784            | 0        |
| 1 | 1.081786    | -1.426218           | 0        |
| 2 | 0.161188    | 0.409875            | 0        |
| 3 | -0.492098   | 0.416183            | 0        |
| 4 | -0.679498   | 0.416183            | 0        |

Figure 1: First 5 rows of tabular data

```
print(df.head())
```

|   | gender | age      | hypertension | heart_disease | smoking_history | bmi       |
|---|--------|----------|--------------|---------------|-----------------|-----------|
| 0 | 0      | 0.01947  | 1.692784     | -0.284439     | 0.936379        | 0.963327  |
| 1 | 0      | 0.01947  | 0.538886     | -0.284439     | -0.282578       | -1.153468 |
| 2 | 1      | 1.187234 | -0.616891    | -0.284439     | -0.282578       | 0.963327  |
| 3 | 0      | 0.01947  | -0.261399    | -0.284439     | -0.282578       | -0.624269 |
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| 4 | -0.679498   | 0.416183            | 0        |

Figure 2: Std, min, max, etc

## Data Loading and Initial Pre-processing — redcode

- Load and Visualize samples:
  - For image data: show one representative image per class label.
  - For tabular data: display example rows and a statistical summary (average, min, max, std for each column).
  - For time-series data: plot a few representative signals.

[h]

- Identify and handle missing data (if applicable for tabular data).

```
import pandas as pd # Load data df = pd.read_csv('data.csv') # Check for missing values
```

- Encode categorical variables and apply feature scaling (if necessary for tabular data).

```
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

Encode categorical columns

```
print(df.head())
```

|   | gender | age      | hypertension | heart_disease | smoking_history | bmi       |
|---|--------|----------|--------------|---------------|-----------------|-----------|
| 0 | 0      | 0.01947  | 1.692784     | -0.284439     | 0.936379        | 0.963327  |
| 1 | 0      | 0.01947  | 0.538886     | -0.284439     | -0.282578       | -1.153468 |
| 2 | 1      | 1.187234 | -0.616891    | -0.284439     | -0.282578       | 0.963327  |
| 3 | 0      | 0.01947  | -0.261399    | -0.284439     | -0.282578       | -0.624269 |
| 4 | 1      | 1.187234 | 1.515858     | 3.515687      | 0.936379        | -0.624269 |

  

|   | HbA1c_level | blood_glucose_level | diabetes |
|---|-------------|---------------------|----------|
| 0 | 1.081786    | 0.847784            | 0        |
| 1 | 1.081786    | -1.426218           | 0        |
| 2 | 0.161188    | 0.409875            | 0        |
| 3 | -0.492098   | 0.416183            | 0        |
| 4 | -0.679498   | 0.416183            | 0        |

Figure 3: First five rows after cleaning (It shows here for some reason)

```
for col in df.select_dtypes(include='object'): df[col] = LabelEncoder().fit_transform(df[col])
```

Feature scaling

```
scaler = StandardScaler() numeric_cols = df.select_dtypes(include=np.number).columns df[numeric_cols] = scaler.fit_transform(df[numeric_cols])
```

- After cleaning, show the same sample rows or plots again to illustrate the improvements (if applicable).

## Motivation and Background

### Refine and formalize your proposal.

Given specific data of a person, such as gender, BMI, blood sugar level, and age among other variables, we can create a prediction of diagnosing diabetes, with a value for 0, negative, and 1, positive. We can see trends among differing sets of people who which help us decide the probability. A study done by Mohamed<sup>2</sup>. can be seen also predicting diabetes using similar datasets that visualize trends in data using bar graphs. Melike Dilekci<sup>3</sup>. is also seen drawing conclusions with a similar dataset but with more relevant variables, such as pregnancies, insulin and skin thickness that creates more precise predictions. [?].

### Provide a rationale for your dataset selection.

With our dataset used for our research, although not as specific, is still helpful and relevant to our subject at hand. Hypertension is helpful to incorporate into our data as they share many symptoms to diabetes. Data variables that are very relevant to diabetes are hbA1c levels, and blood glucose levels, which have a direct relationship to diabetes.

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<sup>2</sup><https://www.kaggle.com/code/mohamedbeshto/diabetes-prediction>

<sup>3</sup><https://www.kaggle.com/code/melikedilekci/diabetes-dataset-for-beginners/notebook>